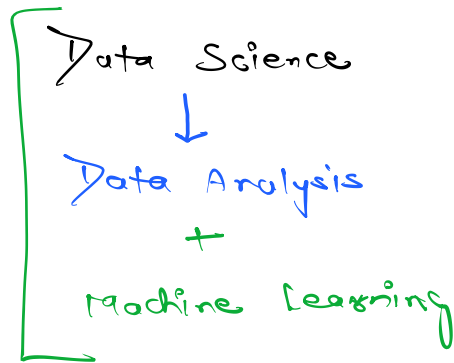




Data Analysis:

- Excel
- SQL
- Tableau/PowerBI
- Statistics
- Python

There are 2 branches of Statistics:

① Descriptive Statistics

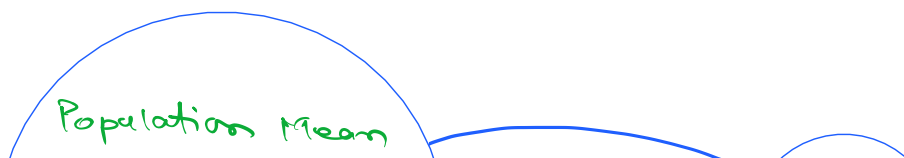
- Measures of Central Tendency
- Measures of Spread/Dispersion

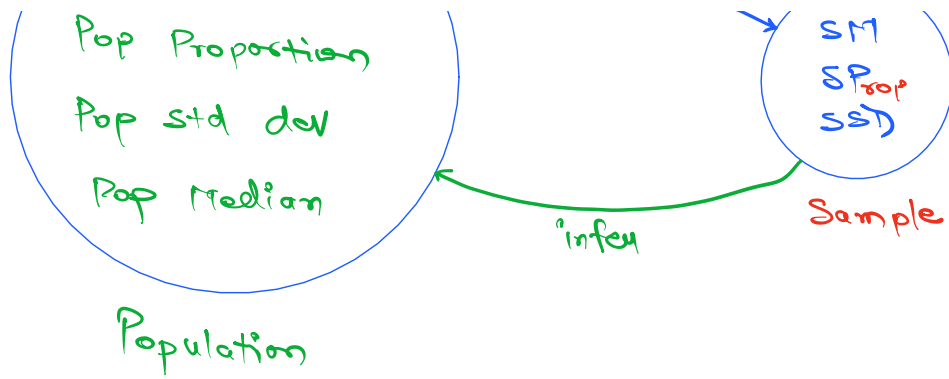
② Inferential Statistics

- Confidence Interval
- Hypothesis Testing
 - A/B Testing

68%

Estimate





Time Constraint

Budget/Resource Constraint

Ex 1: Starbucks → wants to launch a new coffee flavour in India.

Tier 1
Tier 2

↓
Take a sample of customers (5000)

↓
(72%) of them liked it & would order it on regular basis

Ex 2: Spotify:
650 million+ users

PI9 = 42 mins

New PI9: (?)

New song recommendation features

10,000 users (sample)

↓
Observed for 7 days

↓
 $SM = 43.8 \text{ mins}$

—
To infer/estimate population parameters/values using sample values, we have 2 options →

① Point Estimation:

$$SM = 43.8 \text{ mins}$$

(Point Estimate)

1850s

$\text{Pop Mean} = \text{Sample Mean}$

 ✓

② Confidence Interval

$CI = \text{Point Estimate} \pm \text{Margin of Error}$

$$= \text{Sample Mean} \pm MOE$$

$$= 43.8 \text{ mins} \pm 3 \text{ mins}$$

$$= 43.8 - 3 \text{ mins to } 43.8 + 3 \text{ mins}$$

$$CI = \underline{40.8 \text{ mins to } 46.8 \text{ mins}}$$

Pop Mean →

—
Why is it called Confidence Interval?

$$P(\dots) = 0 / (SM \pm MOE)$$

Confidence
Level

90%.

95%.

99%.

Interval (min, max)

43.8 mins $\pm$ 2 mins

43.8 mins $\pm$ 2.5 mins

43.8 mins $\pm$ 3 mins

Formula to calculate Confidence Interval:

$$CI = SM \pm MOE$$

$$= \bar{x} \pm \text{critical value} * \text{Standard Error}$$

$$CI = \bar{x} \pm \left(Z^* * \frac{S}{\sqrt{n}} \right)$$

$\bar{x}$: Sample Mean

Z^* : Z-score/critical value for a certain confidence level.

S : Standard deviation of the sample.

n : Sample size

Questions:

① Minimum sample size?

② Optimal sample size?

③ Sampling Techniques